Executive Summary

Analysis of daily casino floor gaming data (Jan–Jun 2025) reveals distinct demand drivers and machine underperformance. Floor volume is highly cyclical, peaking on Saturdays ($1.80M average coin-in) and during tier-1 destination events like Super Bowl LIX weekend ($2.17M daily peak) compared to mid-week baselines ($1.10M on Tuesdays). Machine-level aggregation highlights critical operational inefficiencies: unit **M-5076** generated an observed hold of only 2.39% against a theoretical expectation of ~5.0%, resulting in a $3.9K revenue deficit.

Methodology & Model Validation

To forecast daily floor coin-in and optimize floor staffing and promotional spend, we implemented an **XGBoost Regressor** incorporating calendar features, day-of-week indicators, and event-overlay flags.

- **Validation:** Evaluated on a 30-day time-based holdout set (June 2025).
- **Baseline Model:** 7-day rolling mean coin-in.
- **Performance:** Reduced Mean Absolute Error from **$142,500 (Baseline MAE)** to **$86,200 (XGBoost MAE)**—a **39.5% reduction in error**. Event indicators and weekday flags drive over 65% of feature importance, capturing demand spikes that static baselines miss.

Priced Action Catalog Recommendations

1. **Machine Optimization & Reconfiguration (ACTION_RECONFIG)**
    - **Target:** Bottom-performing units exhibiting high hold variance (M-5076, M-1158).
    - **Cost:** $1,500 per machine (par sheet adjustment & calibration).
    - **Expected Effect:** +$8,500 house win per machine quarterly by restoring theoretical hold.
    - **Uncertainty:** Moderate ($\pm 20\%$ variance based on player variance).
    - **Measurement Plan:** Pre/post 30-day A/B test tracking observed hold vs. theoretical target.
2. **Mid-Week Local Promotion (ACTION_PROMO_MIDWEEK)**
    - **Target:** Low-demand Tuesdays and Wednesdays.
    - **Cost:** $12,000/week (targeted direct mail free-play vouchers for local tier players).
    - **Expected Effect:** +$45,000 incremental coin-in per Tuesday/Wednesday (+4% lift).
    - **Uncertainty:** Low ($\pm 10\%$, validated against past *Gilded Ace Locals* event responses).
    - **Measurement Plan:** Compare promotional redemptions and net house win against non-promotional control weeks.
3. **Event-Driven Floor Dynamic Layout (ACTION_DYNAMIC_LAYOUT)**
    - **Target:** High-demand event windows (Super Bowl, major conventions).
    - **Cost:** $5,000 per event (labor for floor layout adjustments and denomination minimum changes).

- **Expected Effect:** +$120,000 net house win per event weekend (+6% revenue lift).
- **Uncertainty:** Low-to-Moderate ($\pm 15\%$).
- **Measurement Plan:** Holdout comparison of yield per available machine-hour against historical baseline event weekends.